

# Rayyan Surani

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## EDUCATION

Georgia Institute of Technology, Atlanta, Georgia

December 2027

Bachelor of Science, Computer Science, Concentrations: Intelligence, Modeling & Simulation

GPA: 4.00/4.00

**Relevant Coursework:** Machine Learning, Automata and Complexity, Intro to AI, Design and Analysis of Algorithms, Data Structures and Algorithms, Object and Design

**Honors:** Dean's List (December 2025); Skills USA – Gold Medal, Computer Programming (Georgia, February 2024); Skills USA – Gold Medal, Mathematics (National, June 2023)

## TECHNICAL SKILLS

**Programming Languages:** Java, Python, C++, C, C#, SQL, JavaScript

**Frontend & Backend:** React, Next.js, Vite, Django, Node.js, Flask

**Libraries & Frameworks:** PyTorch, NumPy, Mediapipe, OpenCV, Pandas, Scikit-learn, Hugging Face, LangChain, Spring Boot

**ML/AI – Computer Vision, RAG Systems, LLM Integration**

**Tools & Platforms:** Git, Docker, Linux, AWS (S3, Lambda, EC2, SageMaker), MySQL, PostgreSQL

## WORK EXPERIENCE

### ResMed

Atlanta, GA – Incoming AI Engineer Intern

May 2026 – August 2026

- Developing vertical **AI models using LLMs** in AWS to reduce **clinical documentation time** for healthcare customers
- Building **agentic AI solutions** within a cloud-native microservices architecture using **Java, Spring Boot, and SQL**
- Collaborating with senior engineers and R&D leadership to ship **production-level AI features**

### The Robot Collective – VIP Program

Atlanta, GA – Undergraduate Researcher

January 2026 – Present

- Designing and evaluating **feature expansions** for **swarm robotics systems**, contributing to research on inter-agent communication and emergent behavior strategies
- Training and optimizing **lightweight computer vision models** for real-time swarm robot control, enabling **on-device inference** in constrained hardware environments

## PROGRAMMING PROJECTS

### CareerConnect | AI Career Advisor Chatbot

April 2026

- Architected a **RAG pipeline** grounded in a curated job dataset, enabling the chatbot to surface **semantically relevant listings** with source attribution — improving accuracy over a generic LLM baseline
- Implemented a **FastAPI backend** with **intent-based prompt routing** across four query types (job search, resume feedback, interview prep, Q&A), ensuring contextually grounded LLM responses
- Delivered a **full stack React + Vite application** with an admin dashboard, giving non-technical users direct control over the knowledge base without engineering support

### BiasLens | News Sentiment and Bias Analysis

March 2026

- Engineered a **live article ingestion pipeline** using a custom web scraper, enabling **real-time sourcing and indexing** of news content for continuous analysis
- Built an **end-to-end ML pipeline** that classifies news articles by **political bias and sentiment**, providing transparent, data-driven bias scores instead of subjective impressions
- Deployed a **full-stack web app** supporting user-submitted text and automated article ingestion, making **real-time political bias analysis** accessible through an interactive UI

### Knowledge Recognition Location AI | AI ATL Hackathon

November 2025

- Led **backend and ML engineering** on a 4-person team to build an **auditory aid app** for visually impaired users within a 36-hour hackathon deadline
- Integrated **YOLO object detection** and **MiDaS depth estimation** to enable real-time spatial awareness, warning users of collision risks in their environment
- Implemented **RESTful APIs in Flask** to bridge CV inference results with **Azure LLM and text-to-speech** services, delivering seamless audio feedback to end users

## LEADERSHIP AND COMMUNITY ENGAGEMENT

### Big Data Big Impact, Georgia Institute of Technology

August 2025 – Present

- Participating in **data science workshops** and collaborating on applied projects addressing real-world challenges